

AI-Based Food Waste Prediction and Redistribution Network Using Real-Time Agricultural Data

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Abstract

Food waste has become a major global challenge, with significant quantities of agricultural produce being lost due to inefficient supply chain coordination, poor demand forecasting, and logistical limitations. At the same time, many regions experience food shortages and unequal distribution of resources. This research proposes an AI-based food waste prediction and redistribution network designed to reduce agricultural food waste and improve food distribution efficiency. The proposed system collects real-time data from farms, local markets, storage facilities, and supply chain networks to monitor production levels and market demand. Machine learning models analyze historical and real-time data to predict potential food surpluses and identify areas that may experience shortages. Based on these predictions, the system recommends optimized redistribution strategies to ensure that surplus food can be efficiently transported to regions with higher demand. The framework integrates predictive analytics, real-time monitoring, and intelligent logistics planning to minimize post-harvest losses and improve supply chain coordination. The proposed solution contributes to building a more sustainable and resilient food distribution system that supports global food security.

Keywords— Food Waste Prediction, Supply Chain Optimization, Machine

Learning, Smart Agriculture, Data Analytics, Logistics Planning, Food Distribution

I. INTRODUCTION

Food waste has become one of the most critical challenges within the global agricultural ecosystem, affecting both economic stability and food security. A large portion of agricultural produce is lost after harvest due to inaccurate demand forecasting, inefficient logistics, and poor coordination across supply chains. This inefficiency creates a paradox where significant food surplus exists in some regions, while others continue to face shortages and hunger. Traditional supply chain systems are often built on static planning methods and limited real-time data, making them incapable of responding effectively to dynamic changes in production and demand. As a result, these systems frequently lead to imbalances, causing both excess wastage and unmet demand. With the rapid advancement of artificial intelligence and data analytics, there is a clear opportunity to transform food supply chain management. By leveraging real-time data and predictive modeling, it becomes possible to anticipate fluctuations in supply and demand, enabling more informed and proactive decision-making. In this context, the present study proposes an AI-driven framework designed to predict food waste and support efficient redistribution strategies. The primary objective is to minimize post-harvest losses,

enhance coordination within supply chains, and contribute to the development of more sustainable and balanced food distribution systems.

II. LITERATURE REVIEW

Conventional food supply chain management approaches largely depend on historical data and manual decision-making processes. While these methods offer basic insights, they are inherently limited in their ability to adapt to real-time variations in production, storage conditions, and consumer demand. Recent research has introduced machine learning techniques to improve demand forecasting and supply chain optimization. These models have demonstrated improved predictive accuracy; however, their effectiveness is often constrained by the lack of integrated and diverse data sources. Most existing systems operate in isolation, limiting their ability to provide a comprehensive view of the supply chain. The use of IoT-based monitoring systems has further enhanced data collection by enabling real-time tracking of agricultural production, storage environments, and transportation conditions. Despite this progress, these systems primarily focus on monitoring rather than predictive or decision-oriented capabilities. There is a growing recognition of the need to combine predictive analytics with real-time data to improve food distribution efficiency. However, current research still lacks fully integrated frameworks that not only predict supply-demand imbalances but also translate these predictions into actionable redistribution strategies. This study addresses this gap by proposing a unified, AI-driven system that connects prediction with execution.

III. SYSTEM ARCHITECTURE

The proposed framework is structured around three core components: data collection, predictive analysis, and redistribution planning, each contributing to a seamless and intelligent supply chain process. The data collection layer is responsible for gathering real-time information from multiple sources, including farms, local markets, storage facilities, and transportation networks. This data encompasses key variables such as production volumes, inventory levels, and demand patterns, providing a comprehensive view of the supply chain. The predictive analysis layer processes this data using machine learning models to identify patterns and forecast future supply and demand conditions. By analyzing trends and variations, the system is capable of detecting potential surplus in certain regions and anticipating shortages in others, enabling proactive planning. The redistribution layer translates these predictions into practical action. Based on the identified imbalances, the system generates optimized distribution plans that ensure surplus food is efficiently redirected to areas with higher demand. This not only reduces food wastage but also improves resource utilization and overall supply chain efficiency.

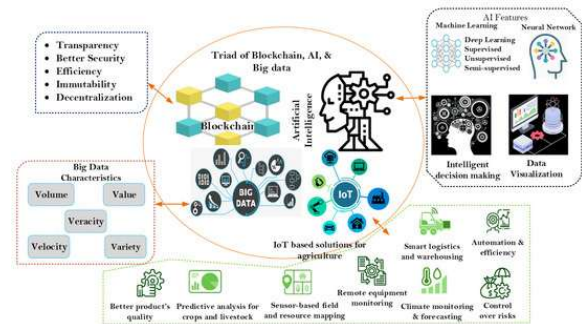


Fig. 1: Overall Architecture of AI-Based Food Waste Prediction System

IV. AI MODEL DESIGN

The predictive component of the proposed system is built using machine learning techniques capable of handling large-scale, heterogeneous datasets. By combining historical records with real-time inputs, the model is designed to capture evolving patterns in production, demand, and supply chain behavior. A crucial step in this process is data preprocessing, which ensures consistency and reliability across multiple data sources. This involves cleaning incomplete or noisy data, normalizing values, and aligning datasets collected from different stages of the supply chain. Feature extraction techniques are then applied to identify the most influential variables affecting supply-demand dynamics, such as seasonal trends, market fluctuations, and transportation constraints. The model is designed to operate in a continuous learning framework, where it updates its predictions as new data becomes available. This allows the system to adapt to changing conditions, improving its forecasting accuracy over time and enabling more responsive decision-making in dynamic supply chain environments.

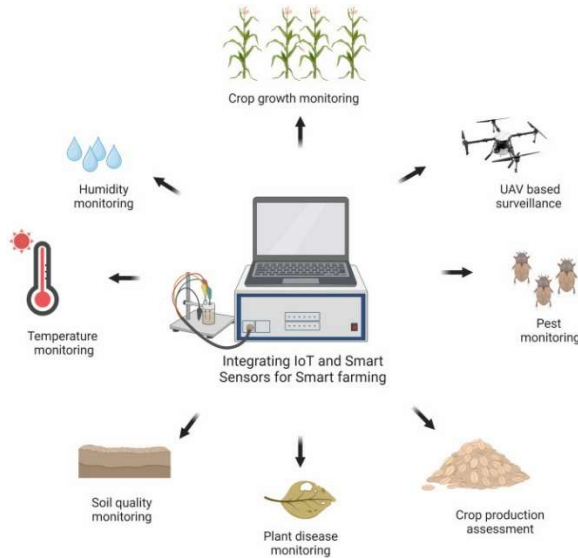


Fig. 2: Data Collection and Integration Framework

V. EXPERIMENTAL SETUP

The proposed system is evaluated using a simulated supply chain environment that reflects real-world agricultural conditions. The dataset incorporates variations in production volumes, market demand, storage conditions, and transportation efficiency to test the robustness of the model under diverse scenarios. The implementation is carried out using Python, supported by machine learning frameworks such as TensorFlow or PyTorch. The dataset is divided into training and testing subsets to ensure an unbiased assessment of model performance. To evaluate effectiveness, multiple performance metrics are considered, including prediction accuracy, reduction in food waste, and the efficiency of redistribution strategies. These metrics provide a balanced view of both predictive performance and practical impact on supply chain optimization.

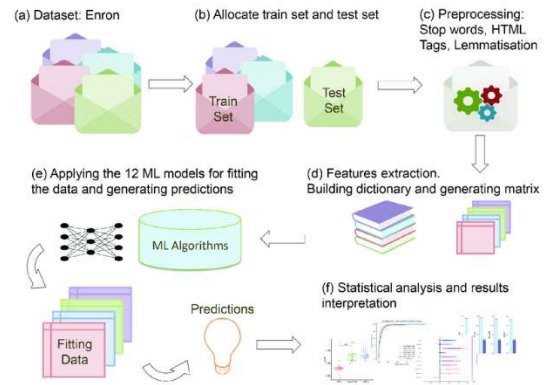


Fig. 3: Machine Learning-Based Prediction Model Workflow

VI. RESULTS AND PERFORMANCE EVALUATION

The experimental results demonstrate that the proposed system significantly enhances food distribution efficiency when compared to traditional supply chain approaches. By accurately predicting supply-demand

imbalances, the model enables early identification of surplus and shortage regions, allowing for proactive intervention. One of the key outcomes is the reduction in post-harvest losses.

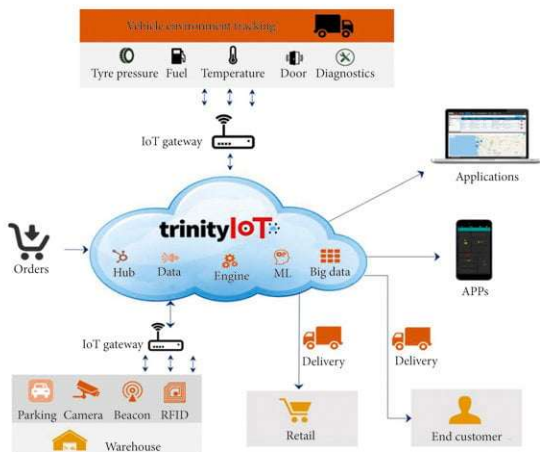


Fig. 4: Food Redistribution Optimization Network

By optimizing redistribution strategies and minimizing delays in transportation and storage, the system ensures that surplus food is utilized effectively rather than wasted. Furthermore, the integration of real-time data improves the reliability and consistency of predictions. Unlike static systems, the proposed framework adapts to changing conditions, maintaining stable performance across different scenarios.

VII. DISCUSSION

The findings emphasize the importance of combining real-time data with predictive analytics to address inefficiencies in food supply chains. The proposed system moves beyond static planning methods by introducing a dynamic and adaptive framework capable of responding to changing conditions in production and demand. However, practical implementation presents several challenges. Reliable access

to high-quality data remains a key concern, as inconsistencies or gaps in data can affect prediction accuracy. Scalability is another important factor, particularly when extending the system to larger geographic regions or more complex supply networks. Additionally, effective coordination among stakeholders—such as farmers, distributors, and market operators—is essential for successful deployment. Without proper collaboration and data sharing, even the most advanced predictive models may fail to achieve their intended impact.

VIII. LIMITATIONS AND FUTURE WORK

The current implementation is based on simulated data, which, although useful for controlled evaluation, does not fully capture the complexity of real-world supply chain environments. In practice, factors such as inconsistent data availability, unpredictable market behavior, infrastructure limitations, and stakeholder coordination challenges can significantly influence system performance. Another limitation is the dependency on data quality and integration. Variations in data formats, missing information, and delays in data collection can affect the accuracy and reliability of predictions. Additionally, scaling the system to handle large, geographically distributed supply chains introduces computational and logistical challenges. Future work will focus on deploying the framework in real-world agricultural supply chains to evaluate its practical effectiveness. Integrating additional data sources—such as weather forecasts, seasonal trends, and transportation analytics—can further improve prediction accuracy and decision-making. Further enhancements may include refining machine learning models for better performance, developing advanced logistics optimization techniques, and exploring hybrid AI

approaches that combine predictive analytics with real-time decision systems. These improvements will help make the framework more robust, scalable, and applicable across diverse agricultural contexts.

IX. CONCLUSION

This study presents an AI-driven framework designed to predict food waste and optimize redistribution using real-time agricultural data. By combining predictive analytics with intelligent supply chain planning, the proposed system addresses key inefficiencies in traditional food distribution methods. The results highlight improvements in supply chain coordination, reduction in post-harvest losses, and more effective utilization of available resources. By enabling proactive decision-making, the framework helps balance supply and demand across regions, contributing to more sustainable and efficient food distribution systems. More importantly, this work demonstrates the potential of data-driven approaches to tackle global food security challenges. With further refinement and real-world implementation, the proposed framework can play a significant role in transforming agricultural supply chains into more adaptive, resilient, and intelligent systems.

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